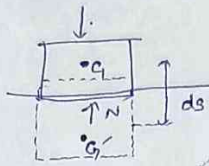


VIRTUAL WORK XD



$ds =$ virtual displacement.

✓ virtual work $= W \times ds$

✓ virtual work $= -N \cdot ds$

$$W \cdot ds - N \cdot ds = 0 \quad \Rightarrow \quad ds \cdot (W - N) = 0$$

$$\Rightarrow \boxed{W = N} \quad [ds \neq 0]$$

Principle of virtual work

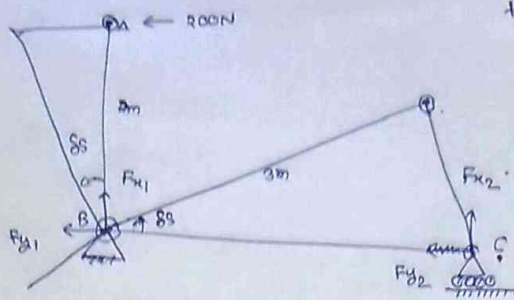
The ^{total} virtual work done of external forces acting on a body under static equilibrium will be 0.

⇒ During the displacement is the user's choice only, he can either rotate etc or whatever he want.

Problem 1

Find the Reaction forces at the joints

metho

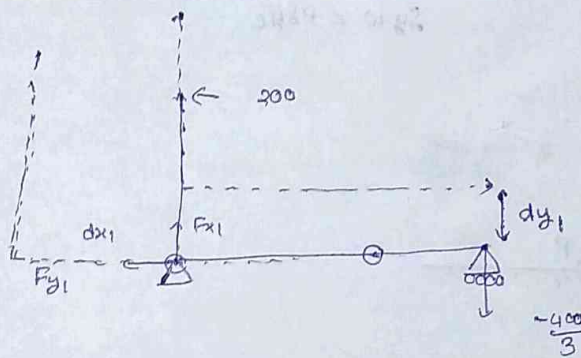


$$[F_{x1} + F_{x2}] dy_2 = 0 \Rightarrow F_{x1} = -F_{x2}$$

$$\text{clearly } = 200 \times 2 \times 80 + R_c \times 3 \times 80 = 0$$

$$\Rightarrow -\frac{200 \times 2}{3} = R_c$$

$$\Rightarrow -\frac{400}{3} = R_c$$

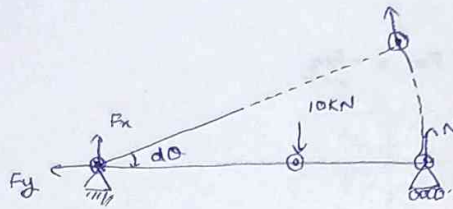
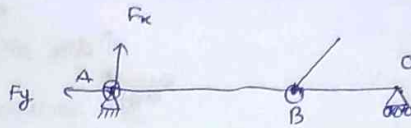


By

$$\sum F_{y1} dx_1 + 200 dx_1 = 0 \Rightarrow F_{y1} = -200 \text{ N}$$

$$-\frac{400}{3} dy_1 + F_{x1} dy_1 = 0 \Rightarrow F_{x1} = \frac{400}{3} \text{ N}$$

Prob 1



$$y_C^2 + x_B^2 = l^2$$

$$2y_C \frac{dy_C}{dx_B} + 2x_B = 0$$

$$\frac{10}{8} \theta d\theta = 3N_R d\theta$$

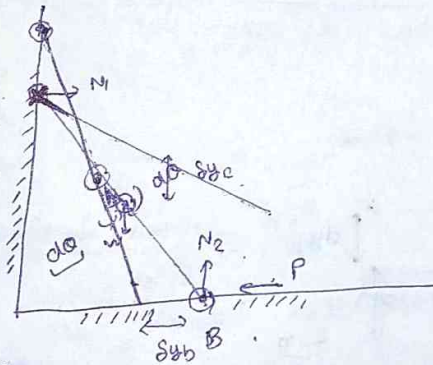
$$\Rightarrow N_C = 5 \text{ kN} \approx 4 \text{ kN}$$

Prob 2

$$\frac{dy_C}{dx_B} = -\frac{x_B}{y_C}$$

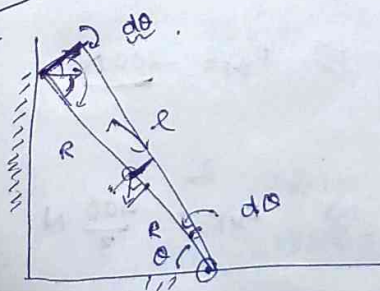
$$\delta y_C = P \delta y_C$$

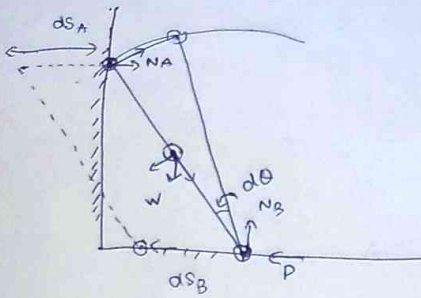
$$\frac{y_C}{x_B} = \frac{\delta y_C}{\delta x_B}$$



$$\frac{\delta y_C}{\delta x_B} = -\frac{x_B}{y_C}$$

$$\frac{\delta y_C}{\delta x_B} = -\frac{1}{2} \cot \theta$$





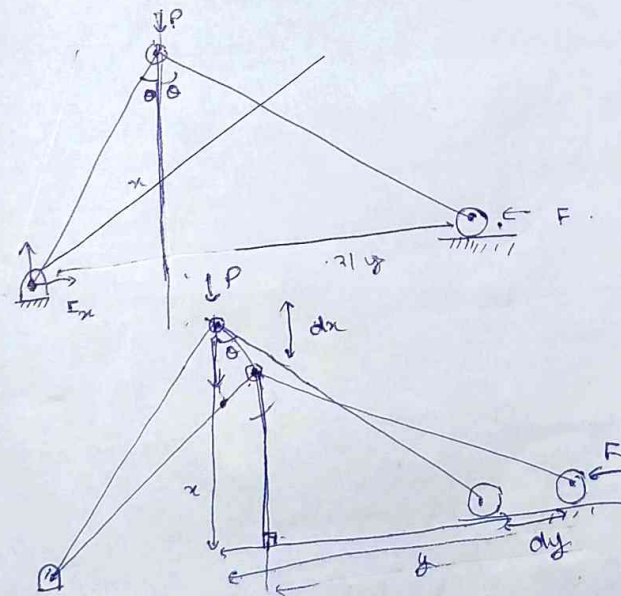
$$ds_A \cdot NA + P ds_B = 0$$

$$P = \frac{NA}{\sin \theta} \quad \text{--- (1)}$$

$$P = \frac{W}{R} \cot \theta$$

**

Prob-3



$$x^2 + y^2 = l^2$$

$$2x dx + 2y dy = 0$$

$$\frac{dy}{dx} = -\frac{x}{y} = \cot \theta$$

virtual

$$P dx - F dy = 0$$

$$P \frac{dx}{dy} = F$$

$$P \tan \theta = F$$